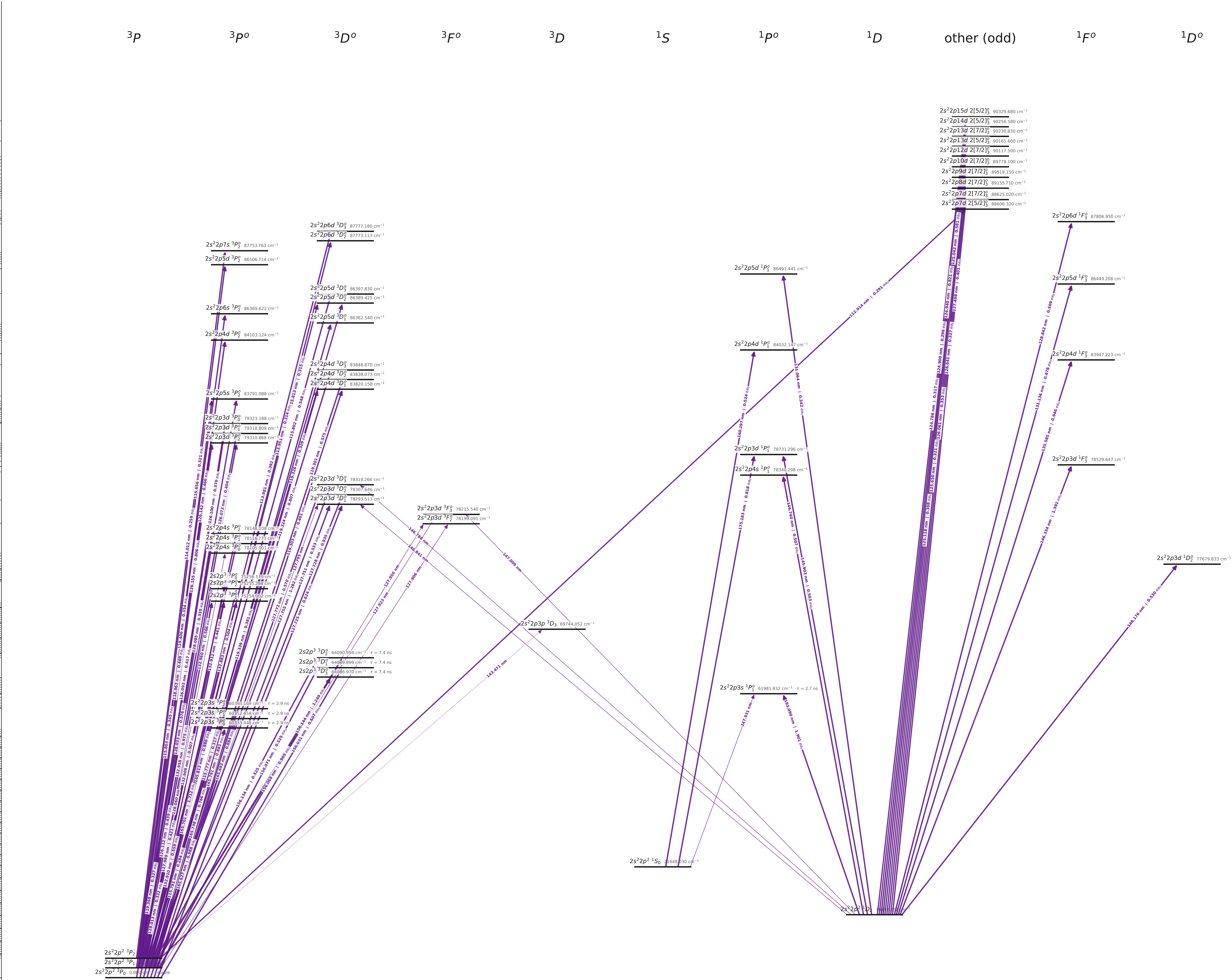


C (C I). Strongest classified transitions below 2  $\mu\text{m}$  (NIST): 59 levels, 80 lines.

$\text{C}^+$  ionisation limit  $90820.35 \text{ cm}^{-1} = 11.26029 \text{ eV}$  (110.107 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $\langle 0 || e r || j \rangle$  in  $e a_0$   
 Matrix elements are measured or NIST values; \* = high-accuracy theory, = model calculation.  
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                          | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $\langle U e U \rangle$ (eao) | A (s <sup>-1</sup> ) | f/2p up. (MHz) | branch (%) | Use                                                           |
|-------------------------------------|--------------------|--------------------|-------------|-------------------------------|----------------------|----------------|------------|---------------------------------------------------------------|
| $2s^22p^2\ ^3P_2 - 2s^22p3p\ ^3D_3$ | 143.4707           | -                  | 2089.5726   | -                             | -                    | -              | -          | two-photon excitation of ground-state carbon atoms at 2 x ... |
| $2s^22p^2\ ^3P_1 - 2s^22p3p\ ^3P_2$ | 165.6267           | -                  | 1810.0495   | 0.988                         | 8.700e+07            | 54.88          | 25.2       | 165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...  |
| $2s^22p^2\ ^3P_0 - 2s^22p3p\ ^3P_1$ | 165.6928           | -                  | 1809.3272   | 0.884                         | 1.160e+08            | 54.88          | 33.6       | 165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...  |
| $2s^22p^2\ ^3P_1 - 2s^22p3p\ ^3P_2$ | 165.7008           | -                  | 1809.2402   | 1.712                         | 2.610e+08            | 54.88          | 75.7       | 165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...  |
| $2s^22p^2\ ^3P_1 - 2s^22p3p\ ^3P_2$ | 165.7379           | -                  | 1808.8350   | 0.766                         | 8.700e+07            | 54.88          | 25.2       | 165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...  |
| $2s^22p^2\ ^3P_1 - 2s^22p3p\ ^3P_2$ | 165.7907           | -                  | 1808.2591   | 0.883                         | 3.470e+08            | 54.88          | 100        | 165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...  |
| $2s^22p^2\ ^3P_2 - 2s^22p3p\ ^3P_1$ | 165.8121           | -                  | 1808.0257   | 0.986                         | 1.440e+08            | 54.88          | 41.8       | 165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...  |
| $2s^22p^2\ ^3P_2 - 2s^22p3p\ ^3P_1$ | 193.0905           | -                  | 1552.6004   | 1.901                         | 3.390e+08            | 58.95          | 91.5       | 193.09 nm line: carbon line used in white-dwarf and comet...  |
| $2s^22p^2\ ^1S_0 - 2s^22p3p\ ^1P_1$ | 247.9310           | 247.8561           | 1209.1770   | -                             | -                    | 58.95          | -          | 247.856 Å                                                     |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

### <sup>13</sup>C hyperfine structure (*I* = 1/2)

| Level               | A (MHz)     | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|---------------------|-------------|---------|--------------------------------------------------------|
| $2s^2 2p^2 \ ^3P_1$ | 2.838(17)   | -       | F=1/2: -2.84 F=3/2: +1.42                              |
| $2s^2 2p^2 \ ^3P_2$ | 149.055(10) | -       | F=3/2: -223.58 F=5/2: +149.06                          |

Measured A, B constants; sources in the data file

## Isotope shifts

| Transition                                 | $\lambda$ vac (nm) | Isotopes | Shift (MHz) |
|--------------------------------------------|--------------------|----------|-------------|
| $2s^2 2p^3 \ ^3P_1 - 2s^2 2p^3s \ ^3P^o_2$ | 165.627            | 13-12    | -3028(1049) |
| $2s^2 2p^3 \ ^3P_0 - 2s^2 2p^3s \ ^3P^o_1$ | 165.693            | 13-12    | -3418(1049) |
| $2s^2 2p^3 \ ^3P_2 - 2s^2 2p^3s \ ^3P^o_2$ | 165.701            | 13-12    | -2908(1049) |
| $2s^2 2p^3 \ ^3P_1 - 2s^2 2p^3s \ ^3P^o_1$ | 165.738            | 13-12    | -2578(1049) |
| $2s^2 2p^3 \ ^3P_2 - 2s^2 2p^3s \ ^3P^o_1$ | 165.791            | 13-12    | -3058(1049) |
| $2s^2 2p^3 \ ^3P_2 - 2s^2 2p^3s \ ^3P^o_1$ | 165.812            | 13-12    | -3118(1049) |
| $2s^2 2p^3 \ ^1S_0 - 2s^2 2p^3s \ ^1P^o_1$ | 247.931            | 13-12    | -4677(60)   |
| $2s^2 2p^3 \ ^1S_0 - 2s^2 2p^3s \ ^1P^o_1$ | 247.931            | 14-12    | -8814(60)   |

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope})$$